

India Rainfall Analytics & Predictive Modeling Dashboard

Comprehensive Hydroclimatic Evaluation (1901–2015), Regional Extremes, Seasonal Monsoon Dependency, and Next-Year Rainfall Forecasting using Linear Regression & Moving Average Models.

Historical Observations: **1901–2015 (115 Years)**

Meteorological Subdivisions: **36 IMD Zones**

District Climatological Normals: **641 Districts**

Data Completeness: **99.86% (70 NA imputed via Long-Term Monthly Normals)**

Target Forecast Year: **2016**

NATIONAL LONG-PERIOD AVERAGE
1,428.17 mm
Baseline annual precipitation normal across India

WETTEST SUBDIVISION (IMD)
ARUNACHAL PRADESH
3,438.47 mm / yr

DRIEST SUBDIVISION (IMD)
WEST RAJASTHAN
292.66 mm / yr

PEAK MONSOON DEPENDENCY
GUJARAT REGION (95.6%)
Lowest: TAMIL NADU (35.1%)

115-YEAR HISTORICAL TRENDS
10 ↑ | 11 ■ | 15 ↓
Total 36 Subdivisions: 42% declining, 28% rising

NEXT-YEAR NATIONAL FORECAST
1,379.6 mm
LPA Departure: -3.4% across India

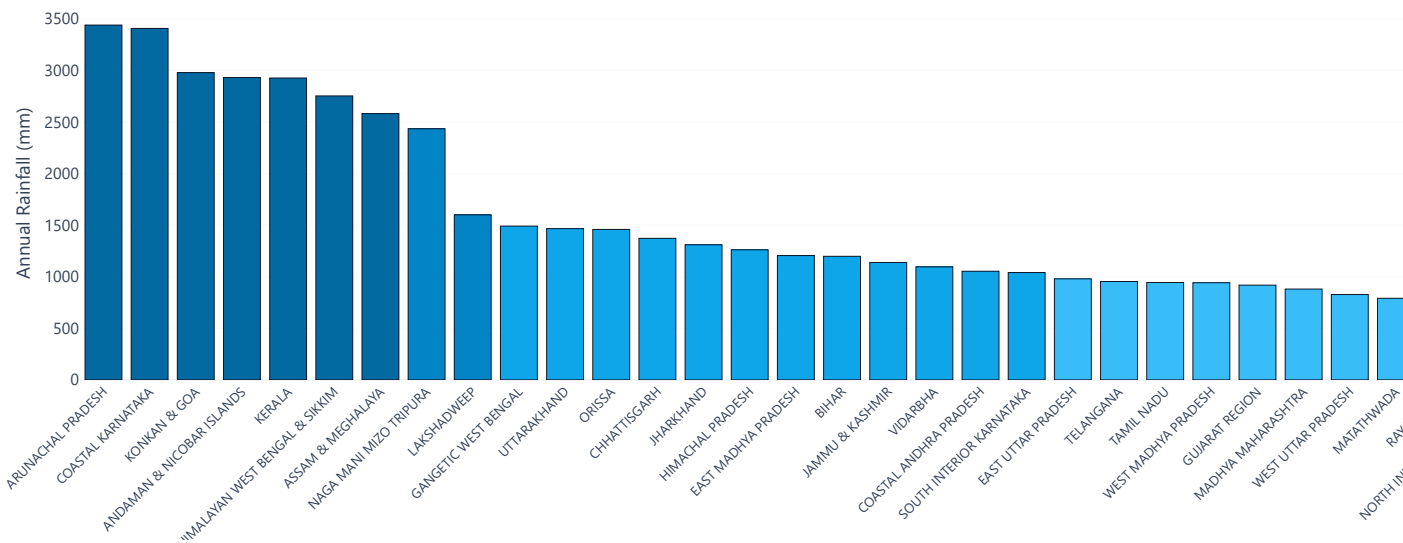
Regional Rainfall Distribution & Extremes

Comparing long-term annual precipitation across Indian subdivisions and states

Bar Chart Ranking

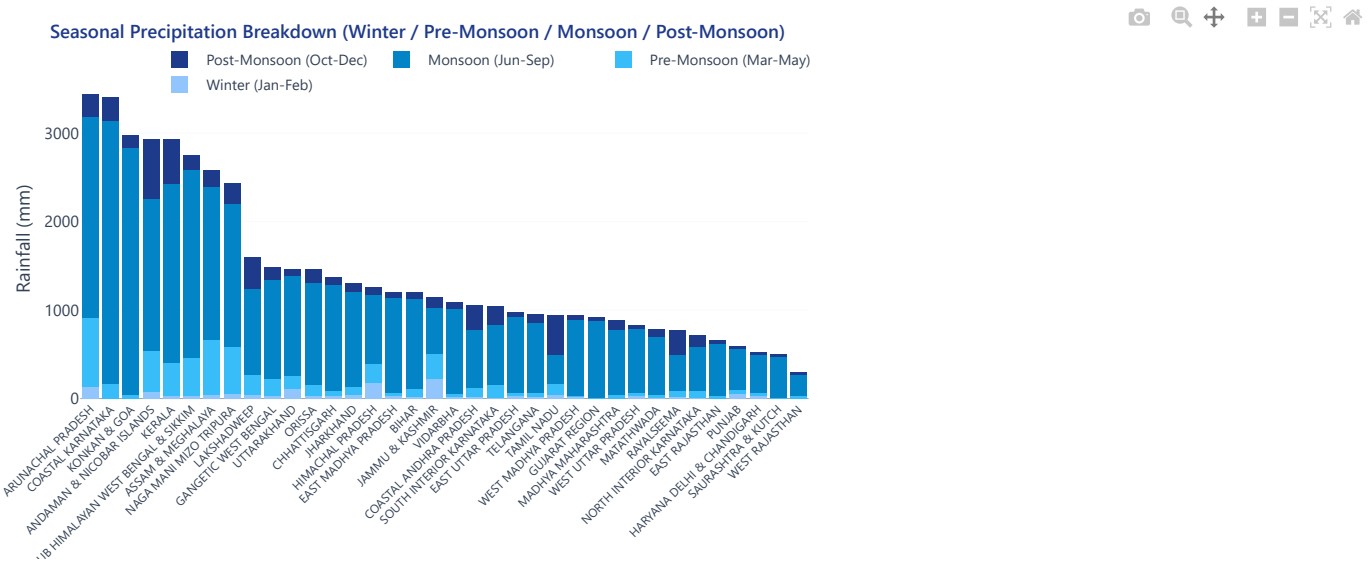
Geographic Bubble Map

Annual Rainfall Normal by Meteorological Subdivision (Sorted Wettest to Driest)



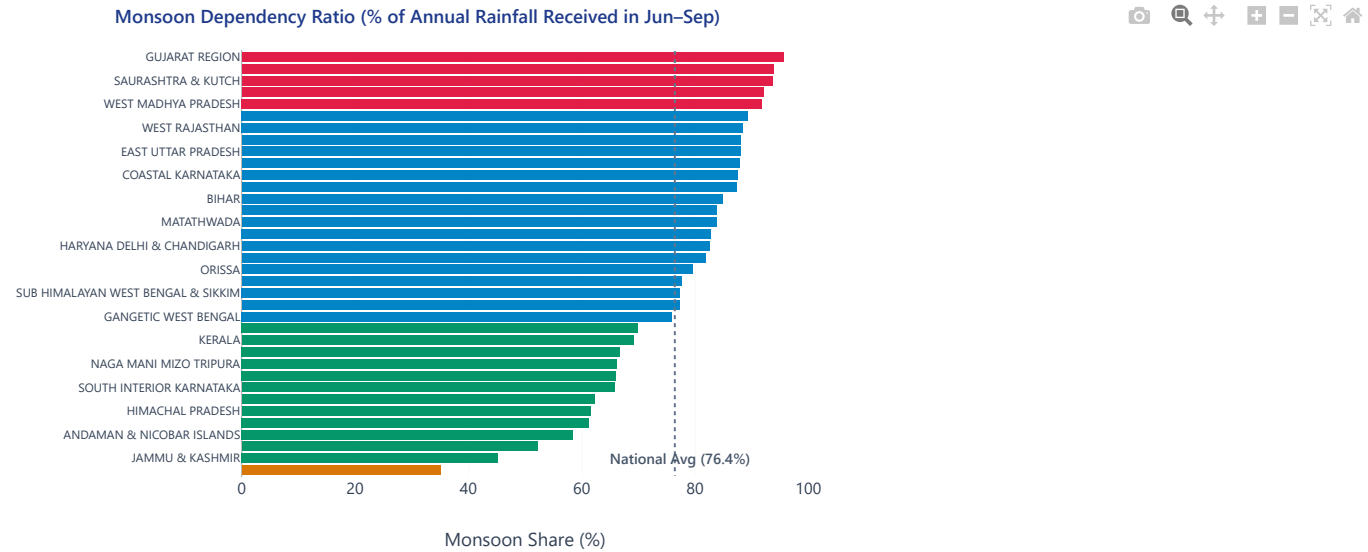
Seasonal Precipitation Composition

Breakdown across Winter (Jan-Feb), Pre-Monsoon (Mar-May), Monsoon (Jun-Sep), and Post-Monsoon (Oct-Dec)



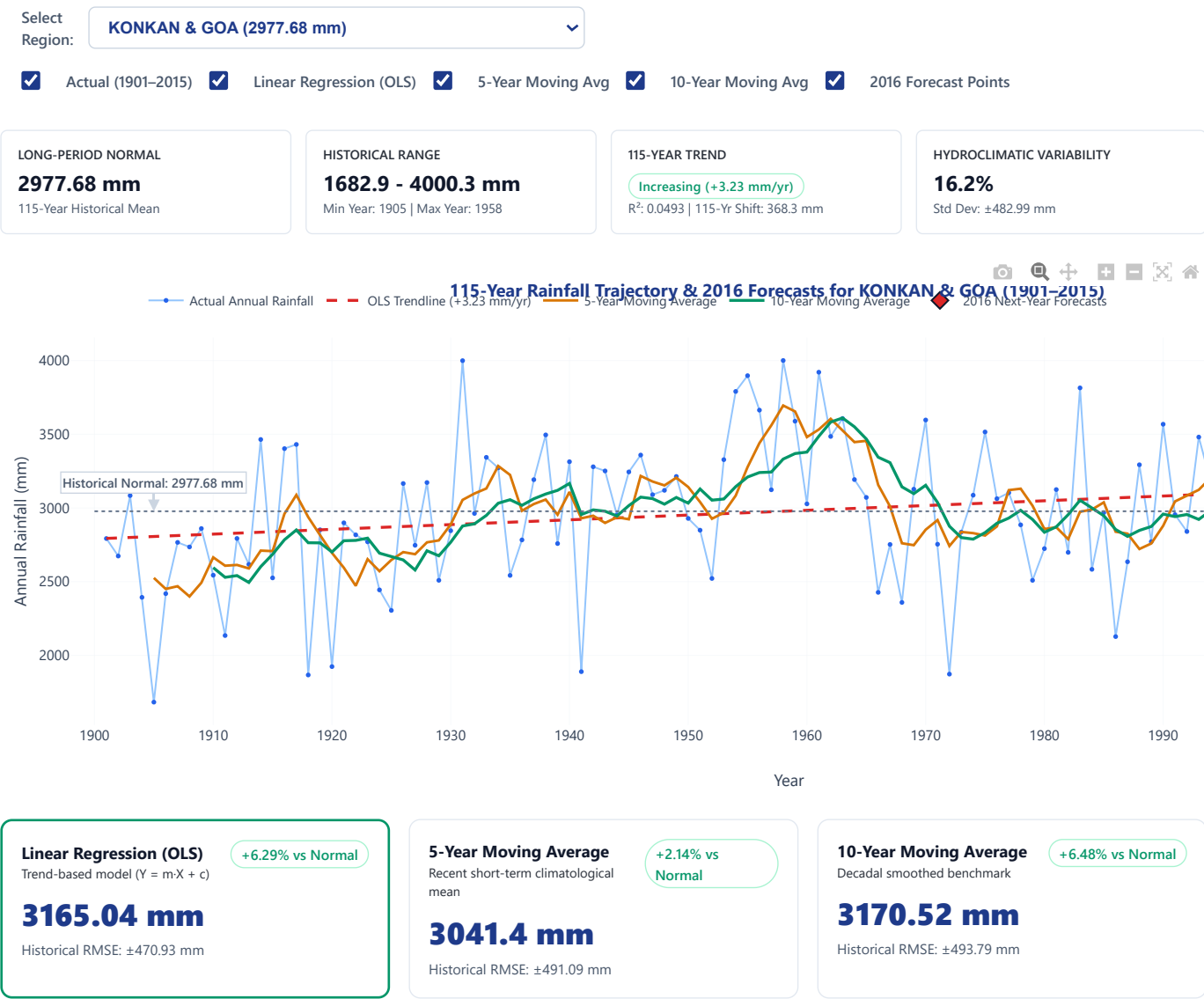
Monsoon Dependency Ratio (%)

Proportion of annual rainfall received during the Southwest Monsoon (Jun-Sep)



Historical Trend & Next-Year Prediction Explorer (1901–2016)

Long-period regression trendline, moving-average smoothers, and next-year forecast evaluation



Complete Regional Forecasts & Statistical Evaluation

Comprehensive data table comparing historical normals, predictive models, trends, and error metrics

Region / Zone 1↓	Historical Normal 1↓	Linear Regression Pred 1↓	Departure (%) 1↓	5-Yr MA Pred 1↓	10-Yr MA Pred 1↓
Arunachal Pradesh	3,438.47 mm	2,192.49 mm	-36.24%	2,569.52 mm	2,497.77 mm
Coastal Karnataka	3,405.92 mm	3,628.87 mm	+6.55%	3,609.73 mm	3,650.13 mm
Konkan & Goa	2,977.68 mm	3,165.04 mm	+6.29%	3,041.4 mm	3,170.52 mm
Andaman & Nicobar Islands	2,930.04 mm	2,856.98 mm	-2.49%	3,322.06 mm	3,083.38 mm
Kerala	2,925.5 mm	2,760.7 mm	-5.63%	2,817.86 mm	2,946.68 mm
Sub Himalayan West Bengal & Sikkim	2,752.2 mm	2,753.67 mm	+0.05%	2,459.64 mm	2,482.6 mm
Assam & Meghalaya	2,580.7 mm	2,432.73 mm	-5.73%	2,169.32 mm	2,261.72 mm
Naga Mani Mizo Tripura	2,433.64 mm	2,179.06 mm	-10.46%	1,778.88 mm	1,908.67 mm
Lakshadweep	1,599.84 mm	1,549.39 mm	-3.15%	1,480.68 mm	1,638.57 mm
Gangetic West Bengal	1,490.49 mm	1,583.15 mm	+6.22%	1,501.98 mm	1,504.7 mm
Uttarakhand	1,465.69 mm	1,384.52 mm	-5.54%	1,411.78 mm	1,379.74 mm
Orissa	1,458.17 mm	1,408.43 mm	-3.41%	1,426.3 mm	1,464.37 mm
Chhattisgarh	1,371.73 mm	1,267.42 mm	-7.6%	1,299.26 mm	1,246.72 mm
Jharkhand	1,309.3 mm	1,210.62 mm	-7.54%	1,186.28 mm	1,184.68 mm
Himachal Pradesh	1,260.34 mm	1,205.05 mm	-4.39%	1,098.26 mm	1,107.43 mm
East Madhya Pradesh	1,205 mm	1,103.31 mm	-8.44%	1,142.9 mm	1,021.93 mm
Bihar	1,197.64 mm	1,124.34 mm	-6.12%	1,026.64 mm	1,050.24 mm
Jammu & Kashmir	1,138.1 mm	1,215.78 mm	+6.83%	1,089.6 mm	1,117.21 mm
Vidarbha	1,095.46 mm	1,073.41 mm	-2.01%	1,086.74 mm	1,079.17 mm
Coastal Andhra Pradesh	1,052.9 mm	1,086.7 mm	+3.21%	1,037.32 mm	1,105.61 mm
South Interior Karnataka	1,040.38 mm	1,102.93 mm	+6.01%	1,074.5 mm	1,117.28 mm
East Uttar Pradesh	979.21 mm	868.04 mm	-11.35%	745.92 mm	735.84 mm
Telangana	953.37 mm	1,022.37 mm	+7.24%	942.8 mm	958.82 mm
Tamil Nadu	943.73 mm	922.95 mm	-2.2%	884.5 mm	964.46 mm
West Madhya Pradesh	941.52 mm	967.78 mm	+2.79%	1,080.46 mm	982.22 mm
Gujarat Region	918.2 mm	934.04 mm	+1.73%	810.26 mm	934.96 mm
Madhya Maharashtra	880.22 mm	952.59 mm	+8.22%	797.5 mm	930.98 mm
West Uttar Pradesh	827.11 mm	772.85 mm	-6.56%	633.84 mm	644.03 mm
Matathwada	790.68 mm	779.62 mm	-1.4%	631.86 mm	707.37 mm
Rayalseema	766.21 mm	798.18 mm	+4.17%	762.98 mm	788.33 mm

REGION / ZONE 11	HISTORICAL NORMAL 11	LINEAR REGRESSION PRED 11	DEPARTURE (%) 11	5-YR MA PRED 11	10-YR MA PRED 11
NORTH INTERIOR KARNATAKA	717.79 mm	739.37 mm	+3.01%	654.18 mm	723.71 mm
EAST RAJASTHAN	655.21 mm	639.95 mm	-2.33%	733.12 mm	673.79 mm
PUNJAB	593.54 mm	586.29 mm	-1.22%	535.78 mm	545.12 mm
HARYANA DELHI & CHANDIGARH	530.49 mm	527.08 mm	-0.64%	421.14 mm	462.93 mm
SAURASHTRA & KUTCH	495.14 mm	559.02 mm	+12.9%	514.66 mm	649.05 mm
WEST RAJASTHAN	292.66 mm	309.81 mm	+5.86%	379.28 mm	352.63 mm

Academic Methodology & Data Processing Notes

1. Data Cleaning & Imputation:

The 1901–2015 dataset contains 4,116 records. Out of 49,392 monthly values, 70 missing entries (0.14%) were detected and imputed using the specific subdivision's long-term monthly mean. Seasonal aggregates (Winter: Jan-Feb, Pre-Monsoon: Mar-May, Monsoon: Jun-Sep, Post-Monsoon: Oct-Dec) and annual totals were mathematically recalculated to ensure exact additive consistency.

4. Climatological Normals & Baseline:

District normals from `district_wise_rainfall_normal.csv` (641 districts across 35 States/UTs) corroborate the sub-divisional long-period averages (LPA), identifying intra-state microclimatic extremes (e.g., Cherrapunji/Mawsynram in East Khasi Hills, Meghalaya vs Jaisalmer, Rajasthan).

2. Trend Analysis Classification:

Trends were determined using Ordinary Least Squares (OLS) linear regression of annual precipitation against year (1901–2015). Regions with absolute slope $|m| \geq 0.5$ mm/year were classified as *Increasing* or *Decreasing*, while those with $|m| < 0.5$ mm/year were classified as *Stable* climatological regimes.

3. Predictive Modeling & Evaluation:

Next-year (2016) forecasts were generated via three methodologies: (1) OLS Linear Regression, (2) 5-Year Moving Average (recent short-term mean), and (3) 10-Year Moving Average (decadal mean). Out-of-sample one-step-ahead backtesting was performed across the historical series to compute Root Mean Squared Error (RMSE) for each model, highlighting the empirical best performer.